



Dr Shashirekha G

Research Associate, IIT Bhilai
Ph.D. in Mathematics (Algebraic Coding Theory)

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- Indian

Research Interests —

- Algebraic Coding Theory
- Finite Fields
- Algebra

Social Network —

- orcid.org/0009-0004-4597-5124
- Google Scholar: Shashirekha G.
- researchgate.net/profile/Shashirekha-G
- linkedin.com/in/shashi888

Memberships —

- Free Graduate Student Member in SAIM from Jan 01 – Dec 12, 2024.
- Life member of Cryptology Research Society of India(CRSI).
- Member of European Women in Mathematics (EWM)

Working Experience

- 2025 – Present **Research Associate** IIT Bhilai
Working in Post Quantum Cryptography.
- 2021 – 2025 **Research Scholar** MIT, Manipal
Department of Mathematics, Manipal Institute of Technology, MAHE, Manipal.
- 2019 – 2021 **Subject Matter Expert (SME)**
Chegg India and Doubtnut.

Education

- 2021 – 2025 **Ph.D. in Mathematics** MIT Manipal, MAHE
Department of Mathematics, Manipal, Karnataka, India.
Thesis Title: “Construction and Structural Properties of Various Types of Cyclic Codes Over Finite Rings.”
- 2017 – 2019 **M.Sc. in Mathematics** University of Mysore
Department of Mathematics, Manasagangothri, Mysore, Karnataka, India.
- 2014 – 2017 **B.Sc. in PMCS** Akkamahadevi Women’s University
Smt. Allum Sumangalamma Memorial College for Women, Ballari, Karnataka, India.
- 2012 – 2014 **PUC** Board of Pre-University
VKJ PU College, Siruguppa, Ballari, Karnataka, India.
- 2011 – 2012 **SSLC** Board of Secondary Education
Shanthinikethana Vidyamandira, Siruguppa, Ballari, Karnataka, India.

Research Papers

- 2026 $\mathbb{Z}_2\mathbb{Z}_2[u]\mathbb{Z}_2[u]$ -Additive Complementary Codes
DMAA
Published a paper titled “ $\mathbb{Z}_2\mathbb{Z}_2[u]\mathbb{Z}_2[u]$ -Additive Complementary Codes” in *Discrete Mathematics, Algorithms and Applications*.
<https://doi.org/10.1142/S1793830926500370>.
- 2026 **Double Cyclic and Quantum Codes**
BSPM
Published a paper titled “*Double Cyclic and Quantum Codes*” in *Boletim da Sociedade Paranaense de Matemática*.
<https://doi.org/10.5269/bspm.78664>.
- 2024 $\mathbb{Z}_2[u]\mathbb{Z}_2[u]$ -Additive Codes
DMAA
Published a paper titled “ $\mathbb{Z}_2[u]\mathbb{Z}_2[u]$ -Additive Codes” in *Discrete Mathematics, Algorithms and Applications*, Vol. 16, No. 08, 2350101.2024.
- 2025 $\mathbb{Z}_2^r\mathbb{Z}_2^s[u]$ -Double Cyclic Codes
In press
G. Shashirekha, G. R. Vadiraja Bhatta, and Prasanna Poojary. Paper titled “ $\mathbb{Z}_2^r\mathbb{Z}_2^s[u]$ -Double Cyclic Codes” has been **accepted** for publication in *Palestine Journal of Mathematics*.
- 2025 **On the Structure and Decomposition of Quasi-Cyclic and Skew-Cyclic Codes**
communicated
G. Shashirekha, G. R. Vadiraja Bhatta, Md Ajaharul Hossain, Ravivarman B., and Saheb Navghare. Work titled “*On the Structure and Decomposition of Quasi-Cyclic and Skew-Cyclic Codes*”. Under review.

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Computer Skills

- MS-Office
- Python
- LATEX

Languages

English	●●●●●
Kannada	●●●●●
Telugu	●●●●●
Hindi	●●●●●

Hobbies

 Playing Chess	
 Listening Music	
 Playing Badminton	

Achievements

2025	AESIM 2025 Summer School on Algebraic Coding Theory, Cryptology and Its Applications <i>Karamanoğlu Mehmetbey University, Karaman, Türkiye</i> Selected to participate in the Summer School on <i>Algebraic Coding Theory, Cryptology and Its Applications</i> , organized by Karamanoğlu Mehmetbey University, Karaman, Türkiye, to be held from June 16–24, 2025.
2025	Construction and Structural Properties of Various Types of Cyclic Codes over Finite Rings <i>MRC 2025</i> Presented a poster titled “ <i>Construction and Structural Properties of Various Types of Cyclic Codes over Finite Rings</i> ” at the <i>Manipal Research Colloquium</i> (MRC 2025), held from April 28–30, 2025, at Dr. T.M.A. Pai Hall, KMC Manipal.
2025	On the Structure of Additive Cyclic Codes <i>Mathematics Research Symposium 2025</i> Presented a paper titled “ <i>On the Structure of Additive Cyclic Codes</i> ” at the 2 nd <i>Mathematics Research Symposium</i> , held at MIT, Manipal, on February 7–8, 2025.
2024	$\mathbb{Z}_2\mathbb{Z}_2(u)\mathbb{Z}_2(u)$-Additive Codes <i>IWATM-2024</i> Presented a paper titled “ <i>$\mathbb{Z}_2\mathbb{Z}_2(u)\mathbb{Z}_2(u)$-Additive Codes</i> ” at the Third International Workshop on Advanced Topics in Mathematics (IWATM-2024), organized by the Centre for Applied Mathematics, IIIT Naya Raipur, held from December 16–20, 2024.
2024	KSET-2024 <i>Karnataka State Eligibility Test</i> Qualified in Mathematics, conducted by the Government of Karnataka.
2023	GATE-2023 <i>Graduate Aptitude Test in Engineering</i> Qualified – Progress in Mathematics.
2023	$\mathbb{Z}_2\mathbb{Z}_2[u]\mathbb{Z}_2[u]$- ADDITIVE CYCLIC CODES <i>ICDMC-2023</i> Presented a paper entitled “ $\mathbb{Z}_2\mathbb{Z}_2[u]\mathbb{Z}_2[u]$ -ADDITIVE CYCLIC CODES ”in International Conference on Discrete Mathematics and Combinatorics at MIT Bengaluru, Karnataka, India from Dec 20-22, 2023
2022	New Linear Codes from Gray Images of Cyclic Codes <i>Ramanujan Mathematical Society</i> Presented paper “ <i>New Linear Codes from Gray Images of Cyclic Codes</i> ” at Ramanujan Mathematical Society’s 37th Annual Conference, SSN College of Engineering, Chennai, Dec 6 – 8, 2022
2016	Quiz Competition <i>National Students’ Union of India (NSUI)</i> Participated in a quiz competition on the 72nd Birth Anniversary of Shri Rajiv Gandhi Ji, organized by NSUI, Ballari, August 20.

Invited Talks

- 2026 **Alumni Talk Series**
MIT, Manipal
Invited to deliver a talk titled “*From MSc to Research: Navigating the Path to PhD and Beyond*”, organized by the School of Basic Sciences, Humanities and Management, Manipal Institute of Technology (MIT), Manipal. Conducted in online mode.

Workshops and conferences

- 2025 **National Instructional Workshop on Cryptology (NIWC 2025)**
Cryptology
Attended the National Instructional Workshop on Cryptology held at IIITDM Kurnool, India from November 24–27.
- 2024 **Applied Number Theory**
CIMPA School
Participated in the CIMPA School on Applied Number Theory held at the University of Education, Ho Chi Minh City, Vietnam from June 3–14.
- 2023 **Finite Geometry and Coding Theory**
CIMPA-NCM School
Participated in the CIMPA–NCM School on Finite Geometry and Coding Theory held at the Indian Institute of Technology Hyderabad, Telangana, India from November 20–December 1.
- 2023 **Advanced Topics in Finite Fields**
ATM School
Attended the Advanced Training in Mathematics workshop on Advanced Topics in Finite Fields, held at IMSc, Chennai, India from July 10–29.
- 2023 **National Instructional Workshop on Cryptology (NIWC 2023)**
Code-based Cryptology
Attended the National Instructional Workshop on Code-based Cryptology held at IIT (BHU), Varanasi, India from July 5–7, gaining insights into code-based techniques.
- 2023 **Algebraic Codes and MAGMA Computation**
IIT Patna
Participated in the online workshop “Algebraic Codes and MAGMA Computation” organized by the Department of Mathematics, IIT Patna, from November 10–11.
- 2021 **Algorithms for Cooperative TU Games**
MIT, Manipal
Registered for the International Workshop on Algorithms for Cooperative TU Games using Matrices and Graphs, held at MIT Manipal, Karnataka, India from December 20–31.
- 2021 **INDOCRYPT – 2021**
LNM Institute of Information Technology
Actively participated in the 22nd International Conference on Cryptology in India held at the LNM Institute of Information Technology, Jaipur during December 13 – 15.
- 2018 **Basic Statistics and Its Application Using SPSS**
Two-Day Workshop
Participated in a workshop on “Basic Statistics and Its Application Using SPSS” organized by the Human Development and Welfare Association and the Nutrition Society, University of Mysore, August 20–21.
- 2017 **Impact of Science and Technology on Society and Economy**
KSTA National Conference
Attended the KSTA National Conference on “Impact of Science and Technology on Society and Economy” jointly organized by the Karnataka Science and Technology Academy (KSTA), Department of Science & Technology, Government of Karnataka, and Vijayanagara Sri Krishnadevaraya University, Ballari, India from March 8–9.

2015 **Bhaskara II & His Works**
One-Day Workshop
Attended a workshop on “Bhaskara II & His Works” organized by Smt. A.S.M. College for Women, and the Indian Association of Physics Teachers, held at Ballari, February 14.

Programs

2025 **Manipal Research Colloquium (MRC)**
MAHE, Manipal
Presented a poster titled “*Construction and Structural Properties of Various Types of Cyclic Codes Over Finite Rings*” in the Mandatory Ph.D. Presentation category during the Manipal Research Colloquium held from April 28–30.

2025 **2nd Mathematics Research Symposium (MaRS 2025)**
MIT, Manipal
Delivered a talk titled “*On the Structure of Additive Cyclic Codes*” at the 2nd Mathematics Research Symposium held at Manipal Institute of Technology, MAHE, from February 7–8.

2024 **Internship**
Internship Program
Participated in a one-month Internship Program under Dr. Ramakrishna Bandi at IIIT Naya Raipur, Chhattisgarh, India, from Jan 27 - Feb 27.

2023 **Scientific Language**
Faculty Development Program
Participated in a one-week Faculty Development Program on “Scientific Language” organized by the Department of Mathematics, School of Basic Sciences, The Assam Kaziranga University, Jorhat, Assam, India, from April 26–May 2.

2023 **Applications of Mathematics in Recent Times**
National Webinar
Participated in the National Webinar on “Applications of Mathematics in Recent Times” held on June 19, organized by the Department of Mathematics, Ghatal Rabindra Satabarsiki Mahavidyalaya, Ghatal, Paschim Medinipur, West Bengal, India.

2022 **Python Programming**
MIT, Manipal
Attended the course “Python Programming” held from June 27–July 3, organized by MAHE–Student Research Forum under the aegis of Directorate of Research, MAHE, Manipal Institute of Technology, Manipal, Karnataka, India.

Certifications & Online Courses

2025 **Cryptography: Learn Public Key Infrastructure from Scratch**
Coursera
Completed an online non-credit course authorized by Packt and offered through Coursera.

2025 **Quantum Computing (One-Month Course)**
CDAC Hyderabad & IIT Roorkee
Completed a one-month course on Quantum Computing organized by the Centre for Development of Advanced Computing, Hyderabad, and the Indian Institute of Technology, Roorkee, with support from the Ministry of Electronics and Information Technology, Government of India.

2025 **Quantum Computing for Everyone – An Introduction**
Coursera
Completed an online non-credit course authorized by Fractal Analytics and offered through Coursera.

2025 **Mathematical Foundations and Quantum Mechanics Essentials**
Coursera
Completed an online non-credit course authorized by Packt and offered through Coursera.

2025	Python Programming for Quantum Computing <i>Coursera</i> Completed an online non-credit course authorized by Packt and offered through Coursera.
2025	Quantum Computing with Qiskit and Advanced Algorithms <i>Coursera</i> Completed an online non-credit course authorized by Packt and offered through Coursera. Covered Qiskit usage and advanced quantum algorithms.
2023	Programming for Everybody (Getting Started with Python) <i>Coursera</i> Completed an online non-credit course authorized by the University of Michigan and taught by Dr. Charles Severance. Introduced Python programming fundamentals.
2022	Cryptography <i>Coursera</i> Completed an online non-credit course authorized by the University of Maryland and taught by Prof. Jonathan Katz. Covered foundational topics in cryptography.